

# Alexandre Ait Ettajer

alexandre.aitettajer@gmail.com  
GitHub: Aiteta

+1(281) 435-9481

Boston, MA  
LinkedIn: alexandre-ait-ettajer

## SKILLS

---

**Languages:** C++, JavaScript, TypeScript, Rust, Go, Python, MATLAB

**Frameworks/Libraries:** MPI, Node.js, LAPACK

**Tools/Platforms:** Docker, GitHub Actions, Git, Perforce, Jenkins

## EXPERIENCE

---

**The MathWorks, Inc.**

Natick, MA (Hybrid)

**Software Engineer**

*April 2025 - Present*

- Enhanced client-server messaging architecture for MATLAB visualizations, allowing internal downstream teams to easily create their own cloud-native visualizations and interactions
- Migrated backend C++ rendering primitives to client-side JavaScript; 25% faster local interactions, 50% faster on MATLAB Online
- Led cross-team initiatives for responsive embedded MATLAB visualizations in web browsers
- Steered working group sessions for distributed visualization architectures across MathWorks teams
- Designed tabular data support for core graphics (Bar, Area, Histogram), integrating SQL/Excel-style datasets
- Resolved mass build/test failures from component dependencies; led Axes/Tick source code restructuring
- Managed cross-cluster Perforce pipelines; contributed to Jenkins production deployments
- Mentored junior engineers; communicated technical direction to senior managers across teams

**Senior Associate Software Engineer**

*April 2024 - April 2025*

- Developed `constantplane` data annotation using MATLAB and JavaScript, allowing users to annotate volumetric data
- Designed and prototyped MATLAB's first hierarchical data viz (Sunburst Chart); led UX and design reviews
- Diagnosed and resolved backend theming and serialization bugs, ensuring compatibility across releases

**Associate Software Engineer / EDG Associate**

*May 2022 - Apr 2024*

- Completed development rotations; gained deep understanding of C++/XML build systems and resolved customer issues

## PROJECTS

---

**Distributed File System** — *Go, Paxos*

Implemented fault-tolerant distributed file system using Paxos consensus for data replication across multiple nodes

**2D Magnetohydrodynamics Simulator** — *C++, CUDA, MATLAB*

Built GPU-accelerated 2D MHD simulator using finite element analysis for fluid simulations

**3D N-Body Simulator** — *C++, MPI*

Developed parallelized gravitational simulator using Barnes-Hut algorithm for  $O(n \log n)$  performance

**Canticle for the Doomed** — *itch.io* — *TypeScript, Docker, GitHub Actions*

Browser-based rogue-like with procedural generation; built hot-reload deployment pipelines (7-day game jam)

## EDUCATION

---

**M.S. in Computer Science**

The University of Texas at Austin — December 2025

*GPA: 3.56, Cum Laude* — Distributed Systems, Parallel Systems, Advanced Operating Systems

**B.S. in Computer Science & B.S. in Aeronautical Engineering**

Rensselaer Polytechnic Institute — May 2022

*GPA: 3.93, Summa Cum Laude*